ABHISHEK HIREMATH

Software Engineer | Backend Engineering | AI & LLM Systems
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PROFESSIONAL SUMMARY

Software Engineer with 1+ years of experience designing and building multi-tenant SaaS backends in Python (FastAPI, Frappe) and TypeScript (Node.js). Strong in REST API design, OAuth 2.0/JWT security, RBAC, and event-driven, queue-based systems (Redis, BullMQ), with PostgreSQL/MariaDB query optimization and caching. Built WhatsApp, Meta, YouTube, and LinkedIn integrations and LangGraph/MCP AI agent services with human approval gates. B.E. in Computer Science and Business Systems (2026).

TECHNICAL SKILLS

Core Competencies: System Design (caching, queues, async processing), REST API Design, Event-Driven Architecture, Multi-Tenant SaaS, AI Agents
Languages: Python, TypeScript, JavaScript (ES6+), Go, SQL, HTML5, CSS3
Backend: Frappe Framework, FastAPI, Node.js, REST APIs, Webhooks, OAuth 2.0, JWT, RBAC
Data & Messaging: PostgreSQL, MariaDB, MySQL, Redis, Drizzle ORM, BullMQ
Cloud & DevOps: Docker, Linux, Nginx, Git, GitHub, GitHub Actions (CI/CD), AWS (Lambda, SQS, SES, SAM)
Testing & Quality: Unit/Integration Testing (Vitest), End-to-End Testing (Playwright), Zod Schema Validation
AI & Integrations: LangGraph, MCP, OpenAI APIs, Prompt Engineering, Meta Graph, WhatsApp Business, YouTube & LinkedIn APIs

PROFESSIONAL EXPERIENCE

Software Engineer Aug 2025 – Present
ZIBMOC Business Solutions Pvt Ltd (walue.biz) Bangalore, India

- Built MSuite, a multi-tenant marketing and CRM platform (Frappe/ERPNext, Python, MariaDB, Vue 3, React) with seven modules (WhatsApp, Social Publishing, Inbox, Automation, Ads, Email, SMS) and realtime UIs, from design through deployment.
- Enforced tenant isolation with workspace-scoped RBAC and plan-based feature gating, and protected client data with OAuth 2.0, JWT propagation, and encryption.
- Integrated five third-party platforms (WhatsApp Business, Facebook, Instagram, YouTube, LinkedIn) through OAuth 2.0 and webhook pipelines, with automated retries for failed message and post deliveries.
- Improved API latency under peak load by profiling slow MariaDB/MySQL queries, adding composite indexes, and introducing Redis caching.
- Designed asynchronous, queue-based workers (Redis, BullMQ) that decouple publishing from request handling and retry failed background jobs.
- Built Campaign Agent, an AI marketing copilot (LangGraph, FastAPI, MCP) that breaks campaign goals into tasks, asks for missing inputs, and requires human approval before publish or spend actions, with streaming events, resumable checkpoints, and audit history.
- Led code reviews, standardized Git branching workflows, and extracted reusable service layers shared across modules.

SELECTED PROJECTS

Sendwell | Multi-Channel Social Scheduling & Messaging Platform (Prototype)
Next.js, React, TypeScript, PostgreSQL, Drizzle ORM, BullMQ, Redis, Socket.IO, Vitest, Playwright

- Designed an event-driven system of publisher, dispatcher, and realtime services; the BullMQ worker refreshes OAuth tokens and publishes to six platforms (LinkedIn, Facebook, Instagram, Threads, X, YouTube).
- Secured the multi-tenant API with OAuth sign-in, workspace-scoped RBAC, hashed API keys, signature-verified inbound webhooks, SSRF-safe outbound webhooks, and AES-256-GCM encryption of stored platform tokens.
- Built WhatsApp, SMS, and email campaigns with audience fan-out and opt-out filtering, plus client review and approval flows and per-platform post analytics.
- Built two-way contact sync with Twenty CRM using signed inbound webhooks, queued outbound updates with echo suppression, a resumable backfill, and sealed credentials.
- Backed by 70+ Vitest test files, Playwright end-to-end tests, Drizzle migrations, and GitHub Actions CI.

Secure MCP Gateway | Agent-to-Platform Integration Layer
TypeScript, Node.js, MCP SDK, Zod, JWT, Redis

- Exposed 36 typed campaign, advertising, social, media, and analytics tools and 5 guided prompts to AI agents over local stdio and authenticated HTTP, forwarding per-tenant JWT identity so every call runs as the caller.

Analytics Agent | Natural-Language Marketing Analytics
Python, FastAPI, MCP, Anthropic / OpenAI / Gemini

- Built a stateless service that answers analytics questions using live MCP tool calls, with a capped tool loop, multi-LLM routing, and per-request caller identity isolated across tenants.

Bulk Messaging Pipeline | Serverless Delivery Plane
TypeScript, AWS SAM, Lambda, SQS, S3, SES, MSG91 / Twilio, Meta Cloud API

- Streamed recipient manifests of 100k+ through queue-driven Lambdas that send email, SMS, and WhatsApp, with retries and a two-lane webhook flow that relays delivery receipts and replies.

EDUCATION

Bachelor of Engineering in Computer Science and Business Systems 2026
Visvesvaraya Technological University (VTU) Karnataka, India